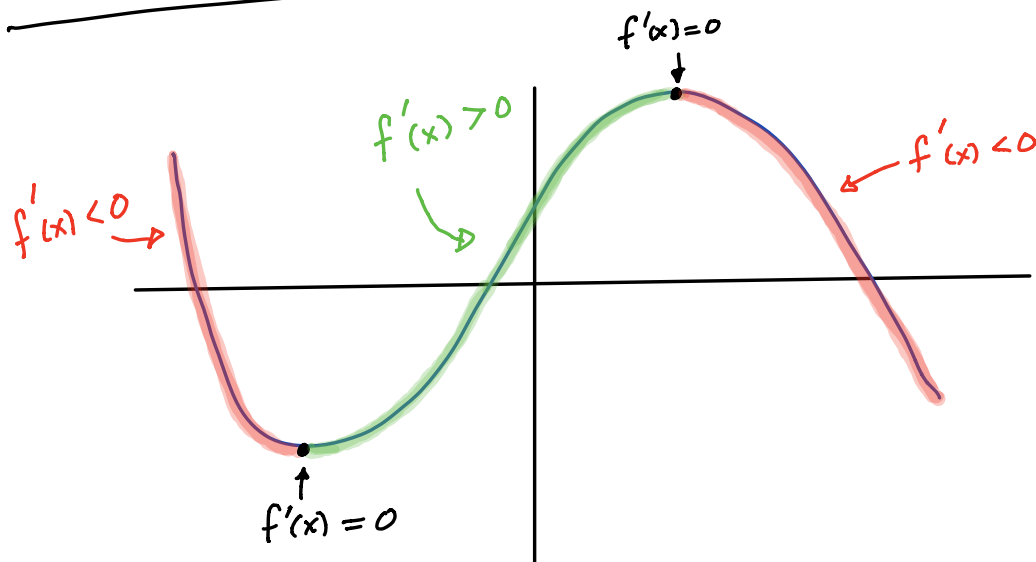
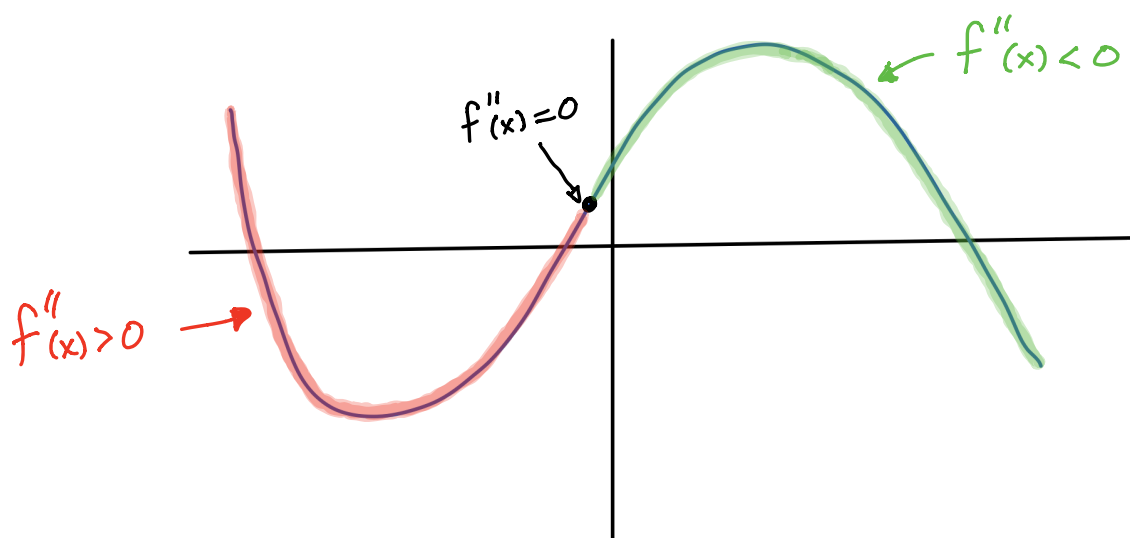


Section 4.3 Derivatives And Graphing



$f(x)$ is increasing where $f'(x) > 0$
 $f(x)$ is decreasing where $f'(x) < 0$

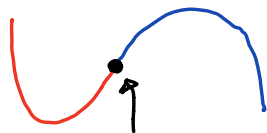
What about the second derivative?



$f(x)$ is concave up where $f''(x) > 0$

$f(x)$ is concave down where $f''(x) < 0$.

The point where $f(x)$ changes from concave up to concave down is called the point of inflection.



inflection point

$$f''(x) = 0 \text{ or } f''(x) \text{ D.N.E.}$$

Ex ① $f(x) = 2x^3 - 3x^2 - 72x + 5$

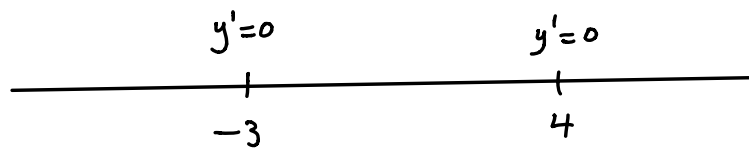
A) Where is $f(x)$ increasing/decreasing?

$$f'(x) = 6x^2 - 6x - 72$$

$$0 = 6(x^2 - x - 12)$$

$$0 = 6(x - 4)(x + 3)$$

Critical Values $x=4$ and $x=-3$

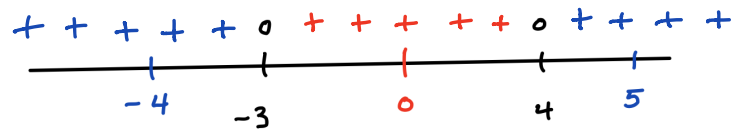


Pick a number in each of the three regions and test each number to see if $f'(x)$ is positive or negative.

Test $x=-2$ $f'(-4) = 6(-4)^2 - 6(-4) - 72 = 48$

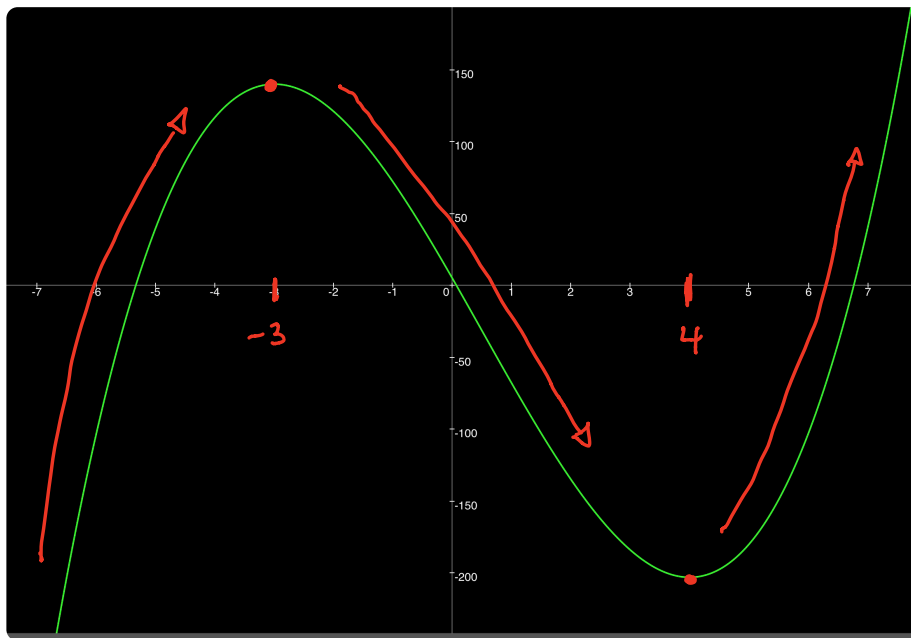
Test $x=0$ $f'(0) = 6(0)^2 - 6(0) - 72 = -72$

Test $x=5$ $f'(5) = 6(5)^2 - 6(5) - 72 = 48$



$f(x)$ increasing $(-\infty, -3) \cup (4, \infty)$

$f(x)$ decreasing $(-3, 4)$



The graph agrees with with
what the math told us about $f(x)$.

B) Where is $f(x)$ concave up/down?

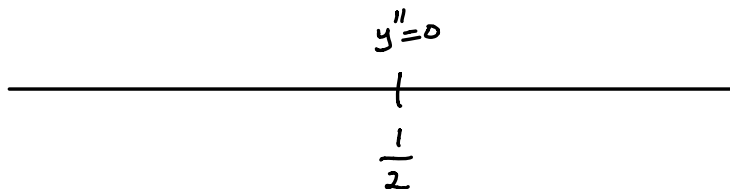
$$f'(x) = 6x^2 - 6x - 72$$

$$f''(x) = 12x - 6$$

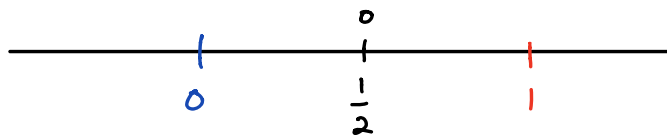
$$0 = 12x - 6$$

$$-12x = -6$$

$$x = \frac{1}{2}$$

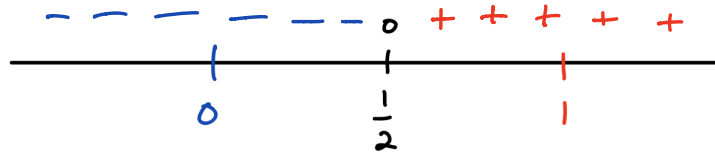


Test each interval to see if $f''(x)$ is positive or negative.



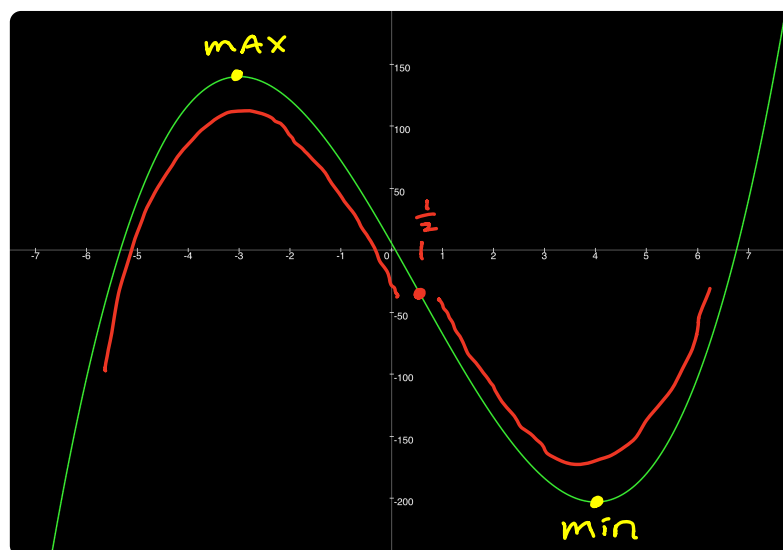
Test $x=0$ $f''(0) = 12(0) - 6 = -6$

Test $x=1$ $f''(0) = 12(1) - 6 = 6$



$f(x)$ concave down $(-\infty, \frac{1}{2})$

$f(x)$ concave up $(\frac{1}{2}, \infty)$



Notice that $f(x)$ is concave down where $f(x)$ has a maximum and $f(x)$ is concave up where $f(x)$ has a minimum.

This is called the Second Derivative Test.

Ex ② Use the Second Derivative Test to

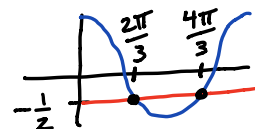
find minimums and maximums for

$$y = x + \sin(2x) \text{ on } [0, \pi].$$

$$y' = 1 + \cos(2x) \cdot 2$$

$$0 = 1 + 2 \cdot \cos(2x)$$

$$\cos(2x) = -\frac{1}{2}$$



$$2x = \cos^{-1}\left(-\frac{1}{2}\right)$$

$$2x = \frac{2\pi}{3} \quad \text{or} \quad 2x = \frac{4\pi}{3}$$

Critical Values $x = \frac{\pi}{3} \quad \text{or} \quad x = \frac{2\pi}{3}$

Now use y'' to find out if there
are maximums or minimums at those
two critical values.

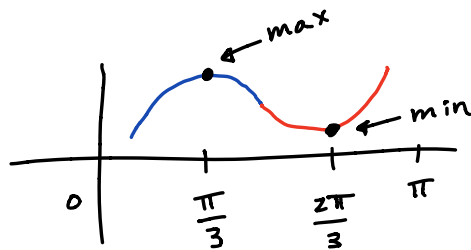
$$y' = 1 + \cos(2x) \cdot 2$$

$$y'' = -\sin(2x) \cdot 4$$

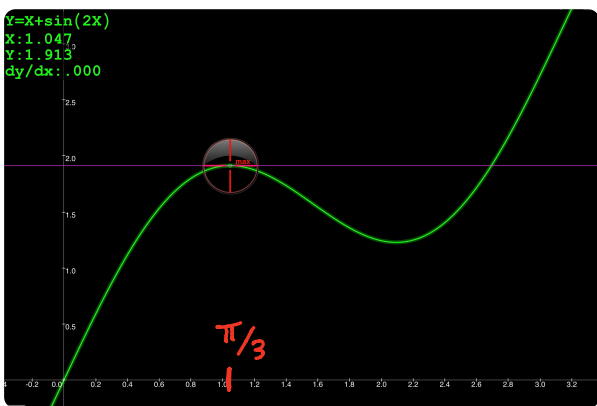
$$y''\left(\frac{\pi}{3}\right) = -4 \cdot \sin\left(\frac{2\pi}{3}\right) = -3.464 \text{ Concave Down}$$

$$y''\left(\frac{2\pi}{3}\right) = -4 \cdot \sin\left(\frac{4\pi}{3}\right) = 3.464 \text{ Concave Up}$$

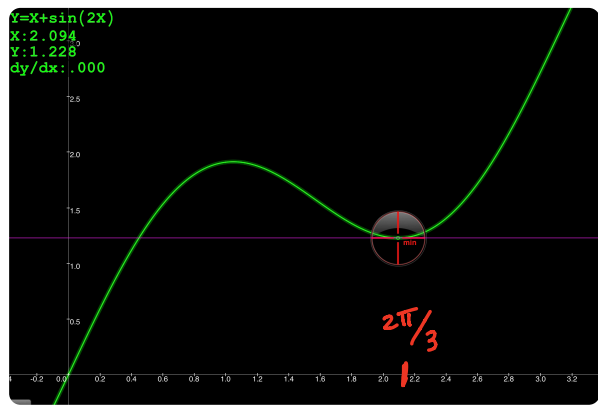
This means that the graph should look something like this.



Now graph $f(x)$ to verify.



Max (1.047, 1.913)



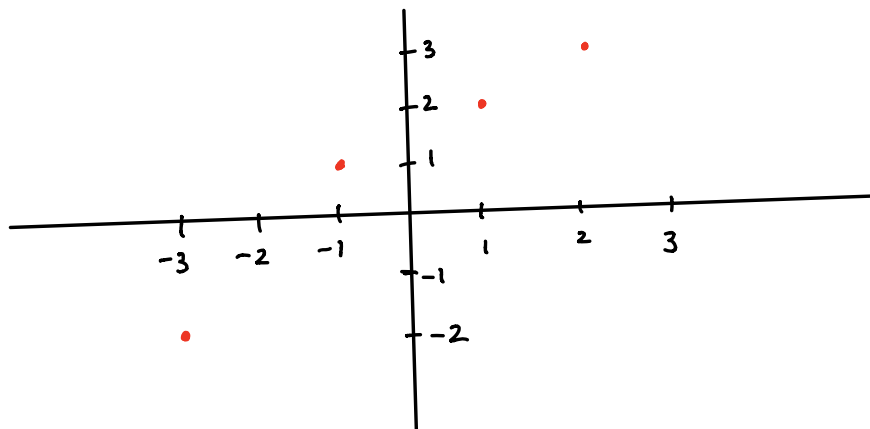
Min (2.094, 1.228)

Ex ③ Draw A continuous function $f(x)$ with All of the following properties.

$$f(-3) = -2, \quad f(-1) = 1, \quad f(1) = 2, \quad f(2) = 3$$

$$f'(-3) \text{ DNE}, \quad f'(-1) = 0, \quad f'(2) = 0$$

$$f'(x) > 0 \text{ only on } (-1, 1)$$

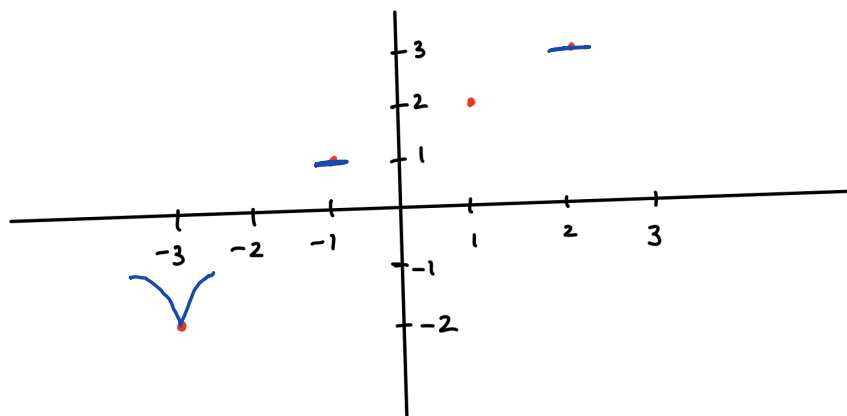


$$f(-3) = -2, \quad f(-1) = 1, \quad f(1) = 2, \quad f(2) = 3$$

Now work with the first derivative.

Put — if $f'(x) = 0$

And use $\vee$ or $\wedge$ if $f' \text{ DNE}$.



$$f'(-3) \text{ DNE}, \quad f'(-1) = 0, \quad f'(2) = 0$$

Now work with $f''(x)$.

$f'(x) > 0$ only on $(-1, 1)$ means that $f(x)$ is concave^U up only between $x = -1$ and $x = 1$; otherwise it is concave[∩] down.

